The Effects of Carbon Loading and Chitosan Coating on Activated Carbon-Alginate Beads for Reactive Textile Dye Removal

Karan Verma, Lezhi Ding, Shantanu Nikam, Arpan Rijal, Joanna Du, Dr. Jessica Ciarla

The University of Texas at Austin
College of Natural Sciences

Background and Objective

Textile dyeing produces large volumes of colored wastewater that can contaminate waterways and harm aquatic ecosystems. **Reactive dyes** are particularly difficult to remove because they are highly soluble in water and resist biodegradation. Powdered **activated carbon** can adsorb these dyes, but recovering the fine particles after treatment is challenging. Previous research has demonstrated that **immobilizing activated carbon** in **calcium alginate beads** allows easier separation, although the surrounding hydrogel may restrict dye access to the carbon.

Chitosan is often added to these composites because its amine groups become protonated under mildly acidic conditions, allowing them to **attract negatively charged dye molecules.**

Objective: Determine how **activated carbon loading** and a **post-gelation chitosan coating** affect **reactive dye removal** and **bead hydration**, individually and in combination.

Higher **carbon loading** was expected to **improve** removal by increasing adsorption sites, while **chitosan coating** was expected to **improve** removal through electrostatic attraction to the **anionic dye.**

Methods



Fig. 1. Experimental workflow for bead preparation, dye treatment, and analysis.

Bead Preparation: A 2 wt% **sodium alginate solution** containing **0, 10, or 20 wt% coconut-shell activated carbon** (relative to alginate solids), was extruded dropwise into 2 wt% **$CaCl_2$** to form crosslinked beads. After gelation, half of each formulation was coated with **chitosan** dissolved in dilute **acetic acid**; the rest remained uncoated. Each of the six formulations was tested in three replicate flasks, giving 18 treatment flasks, plus dye-only controls.

Adsorption: Each flask contained **4.00 g** of blotted wet beads in **40.0 mL** of **Reactive Blue 140** solution (pH 4), shaken at 90 rpm for 24 h.

Measurement: Flasks were photographed in a standardized lightbox under fixed lighting and locked camera settings. Mean **red-channel intensity (I)** was measured using **NIH ImageJ** and converted to image-derived absorbance using the logarithmic form of **Beer–Lambert law**.

$A = -\log_{10}(I/255)$

Apparent dye removal (%) = [(Acontrol − Asample) / Acontrol] × 100

Wet-bead mass gain (%) = [(Final mass − 4.00 g) / 4.00 g] × 100

Statistical Analysis: Two-way ANOVA tested **carbon loading, coating, and their interaction** across 18 flasks. Tukey HSD compared all six formulations; coating contrasts used pooled ANOVA error with Holm-adjusted p-values (a = 0.05).

Post-Treatment Dye Solutions

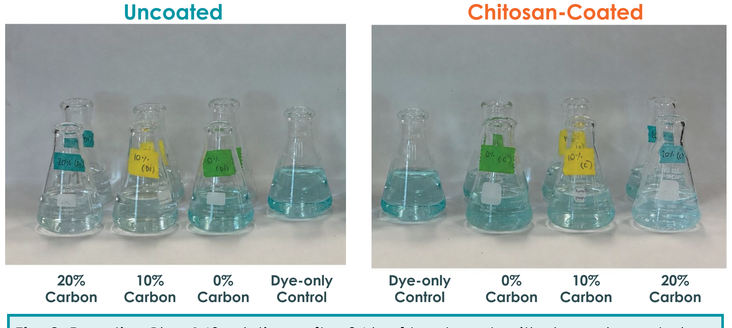


Fig. 2. Reactive Blue 140 solutions after 24 h of treatment, with dye-only controls.

Results

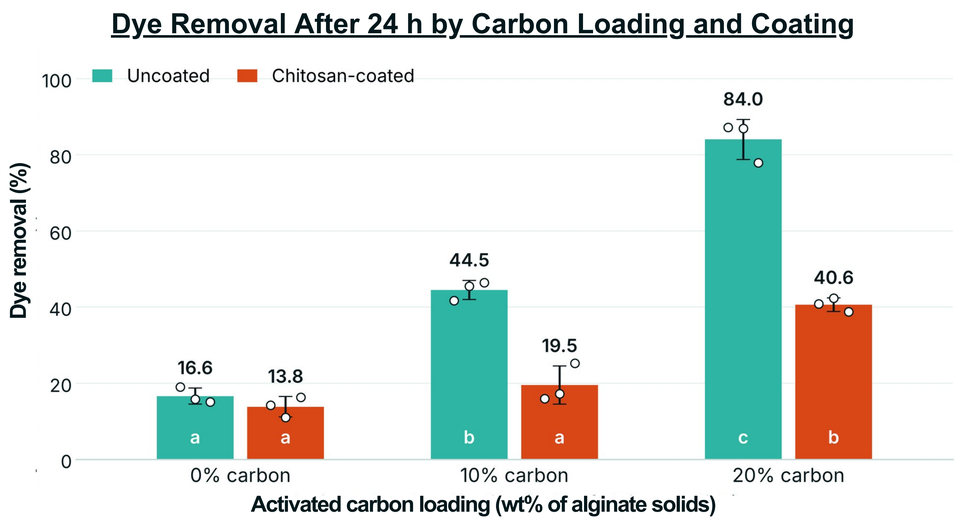


Fig. 3. Dye removal after 24 h (mean ± SD, n = 3 per formulation). Open points show individual flasks; shared letters indicate no significant difference (Tukey HSD, a = 0.05). Two-way ANOVA: carbon loading F(2,12) = 276, coating F(1,12) = 205, interaction F(2,12) = 50.1; all p < 0.001; R^2 = 0.99.

Carbon Loading and Coating Interaction: Increasing carbon loading **improved dye removal** in both formulations, with a substantially greater improvement in **uncoated beads**. The significant interaction indicates that the effect of carbon loading depends on whether the beads are **chitosan-coated.**

Effect of Chitosan Coating: The reduction in **dye removal** associated with coating **increased** from **2.8** percentage points at 0% carbon to **25.0** and **43.3** points at 10% and 20%, respectively. The differences were significant at 10% and 20% carbon (Holm-adjusted p < 0.001), but not at 0% (p = 0.35). The 20% coated and 10% uncoated formulations were not statistically distinguishable (Tukey HSD, adjusted p = 0.76).

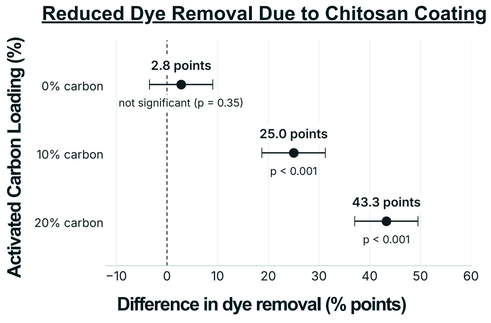


Fig. 4. Difference in dye removal between uncoated and coated beads. Error bars show 95% confidence intervals from the pooled ANOVA error (12 df); p-values are Holm-adjusted.

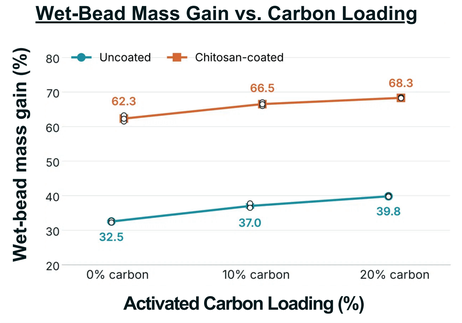
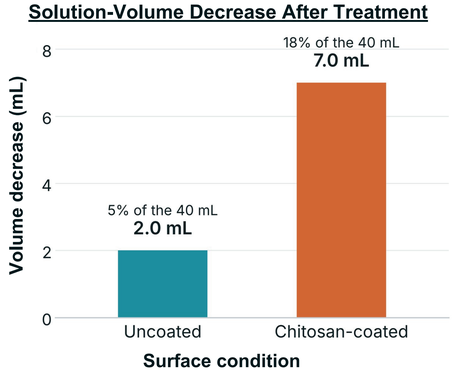


Fig. 5. Wet-bead mass gain relative to the initial 4.00 g wet mass after 24 h. Chitosan coating significantly increased mass gain, while the carbon loading × coating interaction was not significant (p = 0.10).

Effects of Chitosan Coating on Dye Removal and Mass Gain: Coated beads gained **62–68% of their initial wet mass**, compared with **33–40% for uncoated beads**. Although coating increased mass gain by approximately 29 percentage points across all carbon loadings, its **negative effect on dye removal** became progressively greater as carbon loading increased.



Solution Volume and Bead Mass: Flasks with coated beads showed greater solution-volume loss. Alongside **greater wet-bead mass gain** and **lower dye removal**, this suggests that chitosan coating may reduce the beads' **selectivity** for dye adsorption relative to water uptake.

Increased swelling could **restrict dye transport** to the embedded carbon. However, solution-volume loss was not entirely accounted for by bead mass gain, and adsorption selectivity was not directly measured.

Fig. 6. Solution-volume decrease after 24 h for each surface condition, averaged across the three carbon loadings.

Discussion

Carbon Loading Response: Increasing carbon loading from 0% to 20% improved **dye removal by 67.4 percentage points** in uncoated beads, compared with only 26.8 points in coated beads. This suggests that **chitosan coating** limits the effectiveness of additional activated carbon, potentially because increased **water absorption and swelling create a hydrated barrier that restricts dye diffusion to adsorption sites** (Fig. 5). The greater wet-mass gain of coated beads is consistent with this proposed mechanism. Although protonated chitosan can attract anionic dyes at pH 4, this electrostatic advantage may be outweighed by **reduced access to the embedded carbon.**

Proposed Effects of Chitosan Coating on Dye Transport and Absorption

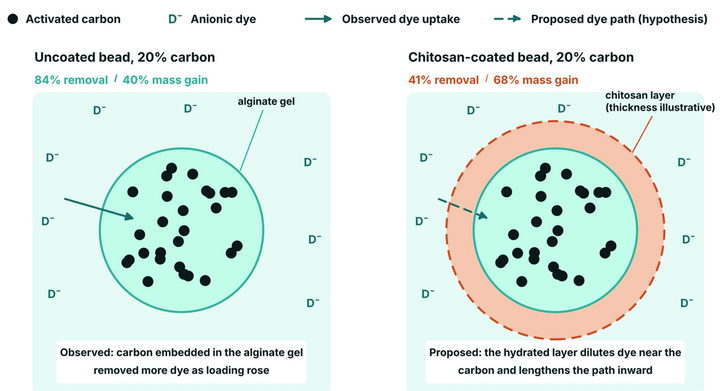


Fig. 7. Proposed mechanism: chitosan coating may increase diffusion resistance and limit dye access to embedded activated carbon, reducing removal despite greater bead hydration.

Conclusions

Increasing **activated carbon** loading substantially improved dye removal. In uncoated beads, **removal rose from 16.6% at 0% carbon to 84.0% at 20% carbon (p < 0.001).**

Chitosan coating **reduced dye removal**, particularly at higher carbon loadings. At 20% carbon, removal **decreased from 84.0% to 40.6%**, with a significant carbon loading × coating interaction (p < 0.001).

Chitosan coating increased wet-mass gain without improving removal. Coated beads gained approximately **1.8 times as much wet mass** and showed **greater solution-volume decreases**, consistent with altered bead–water interactions.

Overall: Uncoated beads containing 20% activated carbon achieved the highest dye removal. The findings highlight the importance of considering dye transport and adsorbent accessibility, rather than electrostatic attraction alone, when designing coated adsorbents.

Limitations and Next Steps: This study tested one type of reactive dye at pH 4 over 24 hours, with three replicates per formulation. Coating thickness, dye transport, and activated-carbon retention were not directly measured.

Future research will focus on optimizing **coating thickness**, testing **higher carbon loadings**, evaluating **bead reuse** over multiple adsorption cycles, and investigating how environmental factors such as salinity and pH affect adsorption. Additional studies could explore methods for **safely destroying or regenerating captured dyes**, including **photocatalytic approaches** using titanium dioxide to degrade collected dye molecules.

References

Annadurai, G., Juang, R.-S., & Lee, D.-J. (2002). Advances in Environmental Research, 6(2), 191–198.

Asadi, S., Eris, S., & Azizian, S. (2018). ACS Omega, 3(11), 15140–15148.

Dutta, S., et al. (2024). Journal of Environmental Management, 353, 120103.

Hermanto, D., et al. (2024). Indonesian Physical Review, 7(3), 480–495.

Saleem, J., et al. (2024). RSC Advances, 14, 33797–33808.

Acknowledgments: We thank **Dr. Jessica Ciarla, Arpan Rijal, and Joanna Du** for their guidance throughout this project.

Scan the QR code to see our website, with the full study, experimental photographs, and supporting data.